Queueing Simulation Part 2 Report

Abstract

To evaluate the optimal number of registers for Mickey's Department Store during the holiday season, M/M/k Queueing theory and MATLAB simulations are employed. The study aims to identify the minimum number of registers necessary to build due to the constraint of ensuring that no more than 10% of customers wait longer than five minutes.

There are two scenarios reviewed in this paper. The first is a 6.5 minute service time per customer. For this scenario, 7 servers are recommended with a probability that a customer in this system will wait more than 5 minutes being 6.46%.

The second is a 8 minute service time per customer when Mickey's Department Store employs trainees. For this scenario, 8 servers are recommended with a probability that a customer in this system will wait more than 5 minutes being 13.54%. However, that poses a contradiction to the overall goal. When working with trainees, it is proposed that 9 servers should be present for 4.57%. Unfortunately, there is only room for 8 serving stations at this store. So, the goal is not possible with trainees.

Introduction
Problem Statement and Solution Methodology

Mickey's Department Store reached out to our firm to prepare for checkout backlog during the holidays. We were asked to decide how many registers are needed to ensure shorter wait times. The first time that I met with the manager, Vicky, I collected the following information about last year's backlog.

Information Gathered from Client	**Parameters for Simulation**
Customer arrival rate: 1 per 1.25 minutes	$\lambda = \frac{1}{1.25} = 0.8$
Rate: 6.5 minutes per customer	$\mu = \frac{1}{6.5} = 0.1538$
Peak period duration: 4 hours or 240 minutes	Run time: 240 minutes
Goal: $\leq 10\%$ of customers wait more than 5 minutes	$P(W_q > 5 \text{ min }) \leq 0.10$
There is room for 8 registers.	$k = 1 : 8$

We were asked to figure out the minimum number of registers that the department store should build to meet the goal of having no more than 10% of customers waiting more than 5 minutes.

First we will calculate the theoretical values. We will narrow down the number of servers that will be most likely to solve our problem using $M/M/k$ queueing theory. The calculations will involve the probability of servers being busy depending on how many customers are currently in the system, number of customers waiting, expected number of customers in the system, expected time customers wait before beginning service, and expected time customers spend in the system. Then we will calculate simulation values for those same items and compare our theoretical values to the simulation values. We will calculate our simulation in minutes.

Theory
M/M/k Queueing Calculations

Now we will review the theory used to solve the problem presented to us one variable at a time to then explain how we will then calculate the required information to solve the problem presented to us.

λ: arrival rate
μ: rate
k: number of servers

$k\mu$: service rate

$k\mu \leq \lambda$ means that the system is not stable, in other words, customers are arriving faster than departing.
$k\mu > \lambda$ means that the system is stable, in other words, customers are departing faster than arriving.

$\rho = \frac{\lambda}{k\mu} < 1$
$\frac{\lambda}{\mu} < k$ gives us how many servers we will need for a stable system.

ρ is the utilization factor, the fraction of the time each serving station is busy. We assume $\rho < 1$ so the queue doesn't diverge.

P_0 represents the probability that the system is empty. P_n is the long-term equilibrium probability of finding n customers in the queue, counting those being served and those waiting. Should $1 \leq n \leq k$, then P_n represents the probability that some cashiers are busy but there is no line. Should $k \leq n$, then P_n represents the probability that all cashiers are busy and customers are either being served or waiting in line. The goal for this problem is to have $max(n) = k - 1$.

$P_n = \begin{cases} \frac{1}{n!}\left(\frac{\lambda}{\mu}\right)^n P_0 & \text{if } 1 \leq n \leq k \\ \frac{1}{k!k^{n-k}}\left(\frac{\lambda}{\mu}\right)^n P_0 & \text{if } k \leq n \end{cases}$ where $P_0 = \frac{1}{\dots}$ since $\frac{1}{P_0} = \sum_{n=0}^{k-1} \frac{1}{n!}\left(\frac{\lambda}{\mu}\right)^n + \frac{1}{k!}\left(\frac{\lambda}{\mu}\right)^k \left(\frac{1}{1-\rho}\right)$

$L_q = \frac{\left(\frac{\lambda}{\mu}\right)^k \rho}{k!(1-\rho)^2} P_0$ is the expected number of customers waiting.

$L = L_q + \frac{\lambda}{\mu} = L_q + s\rho$ is the expected number of customers in the system, including both customers being served and customers waiting.

$W_q = \frac{L_q}{\lambda}$ is the expected time customers wait before beginning service.

$W = \frac{L}{\lambda} = W_q + \frac{1}{\mu}$ is the expected time customers spend in the system, including waiting time and service time.

Theory Application
Theoretical and Simulation
M/M/k Queueing Calculations

From the given information in the first meeting, we have that

arrival rate: $\lambda = \frac{1 \text{ customer}}{1.25 \text{ minutes}} = 0.8$

rate: $\mu = \frac{1 \text{ customer}}{6.5 \text{ minutes}} = 0.1538$

With this information, we will check to see how many servers are required to have a stable system.

$5\mu = 5(0.1538) = 0.769 < .8 = \lambda$ not a stable system
$6\mu = 6(0.1538) = 0.9228 > .8 = \lambda$ system is stable

Since we are told that there is room for 8 serving stations, we will iterate our simulation with consideration of server values $k = 6, 7, 8$. Please note that sometimes texts refer to the number of servers with the variable s as well as k. Since we are looking for the probability that some cashiers are busy but there is no line, we will choose our n values so that $n < k$ for most of them. So, $n = 0, 1, 2, 3, 4, 5, 6$ will only have $k \leq n$ when $k = n = 6$.

$P_0 = \frac{1}{\dots}$ where $\frac{1}{P_0} = \sum_{n=0}^{k-1} \frac{1}{n!}\left(\frac{0.8}{\frac{1}{6.5}}\right)^n + \frac{1}{k!}\left(\frac{0.8}{\frac{1}{6.5}}\right)^k \left(\frac{1}{1-\frac{0.8}{\frac{k}{6.5}}}\right) = \sum_{n=0}^{k-1} \frac{1}{n!}\left(0.8(6.5)\right)^n + \frac{1}{k!}\left(0.8(6.5)\right)^k \left(\frac{k}{k-0.8(6.5)}\right)$

$$P_n = \begin{cases} \frac{1}{n!}\left(0.8(6.5)\right)^n P_0 & \text{if } 1 \le n \le k \\ \frac{1}{k!k^{n-k}}\left(0.8(6.5)\right)^n P_0 & \text{if } k \le n \end{cases}$$

Now we will calculate three sets of values of P_n where the number of customers, $n = 0, 1, 2, 3, 4, 5, 6$, and consider the three possible values of servers that we may have $k = 6, 7, 8$.

n	6 servers P_n
0	0.003204
1	0.016664
2	0.043340
3	0.075146
4	0.097719
5	0.101658

n	7 servers P_n
0	0.004706
1	0.024479
2	0.063665
3	0.110386
4	0.143544
5	0.149331
6	0.129459

n	8 servers P_n
0	0.005226
1	0.027182
2	0.070694
3	0.122572
4	0.159392
5	0.165817
6	0.143751
7	0.106819

$L_q = \frac{(0.8(6.5))^k \frac{0.8(6.5)}{k}}{k!(1-\frac{0.8(6.5)}{k})^2} P_0 = \frac{(0.8(6.5))^{k+1}}{(k-1)!(k-0.8(6.5))^2} P_0$ is the expected number of customers waiting.

$L = L_q + 0.8(6.5)$ is the expected number of customers in the system, including both customers being served and customers waiting.

$W_q = \frac{L_q}{0.8}$ is the expected time customers wait before beginning service.

$W = \frac{L}{0.8}$ is the expected time customers spend in the system, including waiting time and service time.

$P(\text{ error }) = \frac{|\text{ theoretical - simulation}|}{\text{theoretical}}$ is how we calculated the percentage of error

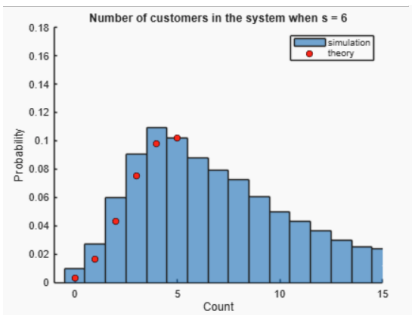
The values for the theoretical are calculated below as expected. From the simulation, we took the mean of all the values generated in our simulation. One of the first things we might notice is that our error percentage is very high. The simulation values for the 4 hour calculation are from *Run_ServiceQueue_baseline_1.m*.The 16 hour calculation is in *Run_ServiceQueue_baseline_1_1.m* where just maxTime variable is changed. Both use 100 samples for the calculation.

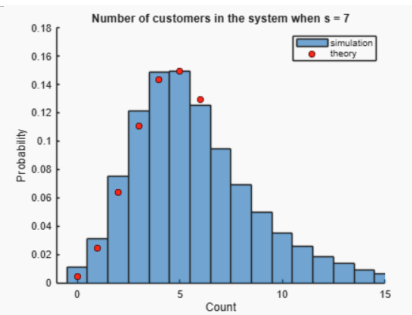
6 servers Value Type	Theoretical	Simulation	Error Percent in 4 hours
L_q	4.3145	2.8982	32.8261
L	9.5160	7.8733	17.2631
W_q	5.3931	3.5360	34.4338
W	11.8950	9.8592	17.1149

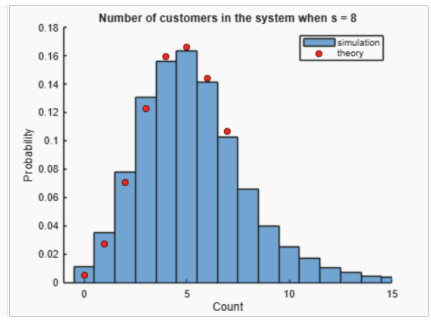
7 servers Value Type	Theoretical	Simulation	Error Percent in 4 hours
L_q	1.0829	0.8263	23.7034
L	6.2845	5.8253	7.3071
W_q	1.3537	1.0131	25.1631
W	7.8556	7.3091	6.9571

8 servers Value Type	Theoretical	Simulation	Error Percent in 4 hours
L_q	0.3690	0.3587	2.7933
L	5.5706	5.4019	3.0281
W_q	0.4613	0.4447	3.5950
W	6.9633	6.7567	2.9666

4 hours simulation graphs shows that the theoretical P_n values (red dots) differ from the simulation probabilities (blue bars) significantly.







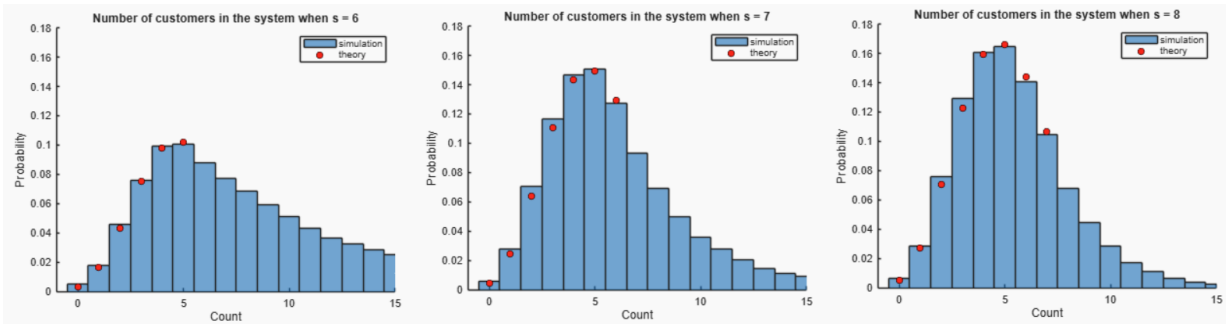
When we run the simulation for 8 hours instead of 4 hours, we get a smaller margin of error. For instance, if we run the simulation for 16 hours, the error percentage is much smaller. This is because the theoretical values are for calculations over the long run (a longer period of time than what we did here).

6 servers Value Type	Theoretical	Simulation	Error Percent in 16 hours
L_q	4.3145	3.9848	7.6400
L	9.5160	9.1555	3.7886
W_q	5.3931	4.9674	7.8929
W	11.8950	11.4395	3.8298

7 servers Value Type	Theoretical	Simulation	Error Percent in 16 hours
L_q	1.0829	1.0053	7.1728
L	6.2845	6.1115	2.7524
W_q	1.3537	1.2579	7.0743
W	7.8556	7.6821	2.2092

8 servers Value Type	Theoretical	Simulation	Error Percent in 16 hours
L_q	0.3690	0.3302	10.5321
L	5.5706	5.4497	2.1708
W_q	0.4613	0.4156	9.9184
W	6.9633	6.8508	1.6150

16 hours simulation graphs shows that the theoretical P_n values (red dots) differ much less now from the simulation probabilities (blue bars) than the simulation with just 4 hours.



Theory Application
Evaluating W_q Calculations

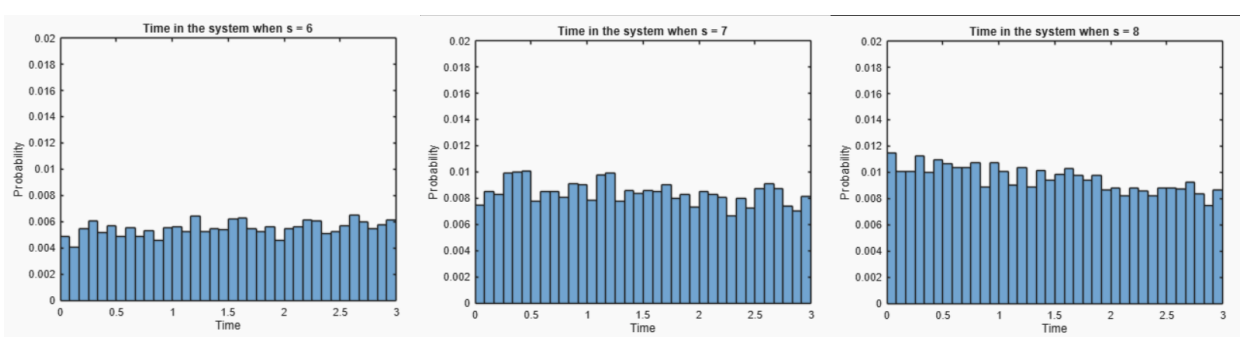
Now back to calculations in a simulation that lasts only 4 hours. The W_q calculations represent the expected time customers wait before beginning service. We were asked to figure out the minimum number of registers that the department store should build to meet the goal of having no more than 10% of customers waiting more than 5 minutes. To do this, we will review the applicable simulation values shown in the chart below.

servers	Theoretical	Simulation	Total Customers	# of $W_q > 5$ min	$P(W_q > 5)$	$P(W_q > 5) \leq 0.10$?
6	5.3931	3.5360	18370	5053	0.27507	No > 10%
7	1.3537	1.0131	18532	1197	0.064591	Yes < 10%
8	0.4613	0.4447	18674	474	0.025383	Yes < 10%

$P(W_q > 5) = \frac{\text{\# of } W_q > 5 \text{ min}}{\text{Total Customers}}$ is the probability that the expected time customers wait before beginning service is greater than 5 minutes

The minimum number of servers recommended to meet the goal is $s = 7$ since
$P(W_q > 5) = \frac{1197}{18532} = 0.064591 < 10\%$.

Please note that these numbers come from a simulation with 100 Samples. Since $7 < 8$ and there is room for 8 registers, there will be room for 7 registers. Here is a graph showing the probability that some cashiers are busy but there is no line during each of these scenarios.



Plot Twist
Problem Statement and Solution Methodology

When meeting with another member of Mickey's Department Store, the Human Resources Director Jackie, I was informed that due to several recent hires, the new average service time might be closer to 8 minutes.

Information Gathered from Client	**Parameters for Simulation**
Customer arrival rate: 1 per 1.25 minutes	$\lambda = \frac{1}{1.25} = 0.8$
Rate: 8 minutes per customer	$\mu = \frac{1}{8} = 0.125$
Peak period duration: 4 hours or 240 minutes	Run time: 240 minutes
Goal: $\leq 10\%$ of customers wait more than 5 minutes	$P(W_q > 5 \text{ min }) \leq 0.10$
There is room for 8 registers.	$k = 1:8$

We were asked to figure out the minimum number of registers that the department store should build to meet the goal of having no more than 10% of customers waiting more than 5 minutes. So we shall edit the code for the simulation by changing just one value, μ, and recalculate the values.

Theory Application
Theoretical and Simulation
M/M/k Queueing Calculations for Updated μ value

The only value that changes in our calculation is $\mu = \frac{1}{8} = 0.125$. Here are the updated formulas. You may find the code for this in *Run_ServiceQueue_sensitivity_1.m*.

Recall this from earlier:
$k\mu \leq \lambda$ means that the system is not stable, in other words, customers are arriving faster than departing.
$k\mu > \lambda$ means that the system is stable, in other words, customers are departing faster than arriving

$5\mu = 5(0.125) = 0.625 < .8 = \lambda$ not a stable system
$6\mu = 6(0.125) = 0.75 < .8 = \lambda$ not a stable system
$7\mu = 7(0.125) = 0.875 > .8 = \lambda$ system is stable

This is why we will iterate our simulation loop for k = 7:8.

$$P_0 = \frac{1}{\dots} \text{ where } \frac{1}{P_0} = \sum_{n=0}^{k-1} \frac{1}{n!}\left(\frac{0.8}{\frac{1}{8}}\right)^n + \frac{1}{k!}\left(\frac{0.8}{\frac{1}{8}}\right)^k \left(\frac{1}{1-\frac{0.8}{\frac{k}{8}}}\right) = \sum_{n=0}^{k-1} \frac{1}{n!}(0.8(8))^n + \frac{1}{k!}(0.8(8))^k \left(\frac{k}{k-0.8(8)}\right)$$

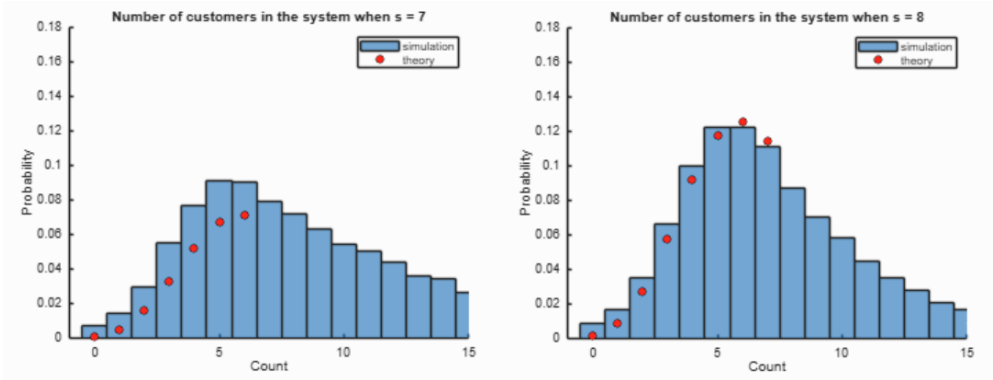
$$P_n = \begin{cases} \frac{1}{n!}(0.8(8))^n P_0 & \text{if } 1 \leq n \leq k \\ \frac{1}{k!k^{n-k}}(0.8(8))^n P_0 & \text{if } k \leq n \end{cases}$$

Now we will calculate three sets of values of P_n where the number of customers, $n = 0, 1, 2, 3, 4, 5, 6$, and consider the three possible values of servers that we may have $k = 7, 8$.

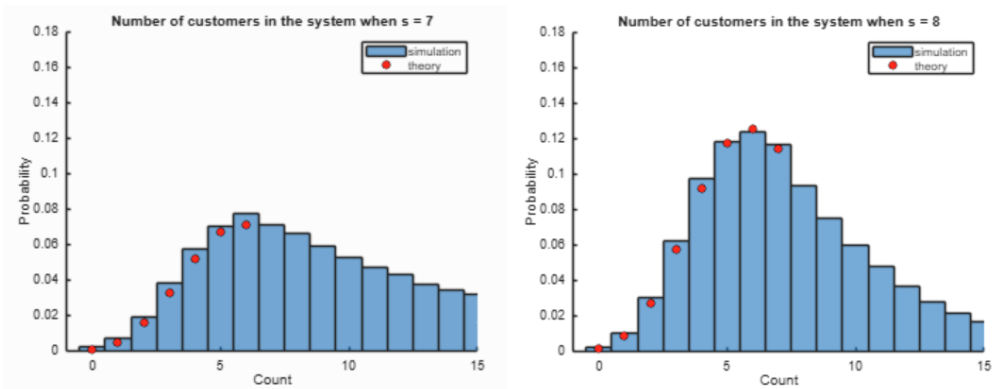
n	7 servers P_n
0	0.000744
1	0.004760
2	0.015233
3	0.032497
4	0.051995
5	0.066553
6	0.070990

n	8 servers P_n
0	0.001311
1	0.008391
2	0.026852
3	0.057283
4	0.091653
5	0.117316
6	0.125137
7	0.114411

4 hours simulation graphs shows that the theoretical P_n values (red dots) differ from the simulation probabilities (blue bars) significantly.



16 hours simulation as shown in *Run_ServiceQueue_sensitivity_1_1.m* gives us more of a parallel with simulation and theoretical values in the graph.



$L_q = \frac{(0.8(8))^k \frac{0.8(8)}{k}}{k!(1-\frac{0.8(8)}{k})^2} P_0 = \frac{(0.8(8))^{k+1}}{(k-1)!(k-0.8(8))^2} P_0$ is the expected number of customers waiting.

$L = L_q + 0.8(8)$ is the expected number of customers in the system, including both customers being served and customers waiting.

$W_q = \frac{L_q}{0.8}$ is the expected time customers wait before beginning service.

$W = \frac{L}{0.8}$ is the expected time customers spend in the system, including waiting time and service time.

$P(\text{ error }) = \frac{|\text{ theoretical - simulation}|}{\text{theoretical}}$ is how we calculated the percentage of error

7 servers Value Type	Theoretical	Simulation	Error Percent in 4 hours
L_q	8.0771	3.9535	51.0535
L	14.4771	9.9523	31.2548
W_q	10.0964	4.6884	53.5632
W	18.0964	12.4112	31.4163

8 servers Value Type	Theoretical	Simulation	Error Percent in 4 hours
L_q	1.8306	1.6236	11.3081
L	8.2306	7.7793	5.4826
W_q	2.2882	1.9761	13.6421
W	10.2882	9.7062	5.6574

Run_ServiceQueue_sensitivity_1.m is the 4 hour simulation while *Run_ServiceQueue_sensitivity_1_1.m* is the 16 hour simulation. You can see that the 16 hour simulation has lower margins of error between the theoretical and simulation values for 7 servers. However for 8 servers, not much changed with respect to the margin of error.

7 servers Value Type	Theoretical	Simulation	Error Percent in 16 hours
L_q	8.0771	6.2825	22.2189
L	14.4771	12.5947	13.026
W_q	10.0964	7.7801	22.9417
W	18.0964	15.7365	13.0406

8 servers Value Type	Theoretical	Simulation	Error Percent in 16 hours
L_q	1.8306	1.5991	12.6431
L	8.2306	7.9145	3.8405
W_q	2.2882	1.9936	12.8755
W	10.2882	9.9174	3.6042

Theory Application
Evaluating W_q Calculations for the Updated μ

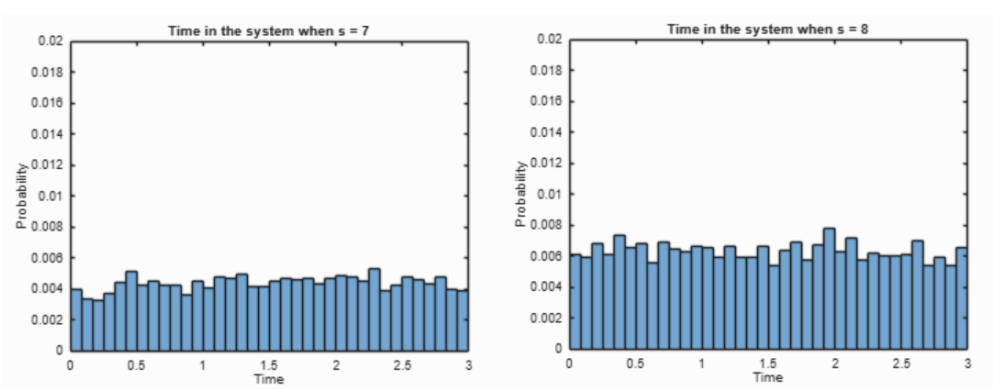
Now we are back to the 4 hour simulation.

servers	Theoretical	Simulation	Total Customers	# of $W_q > 5$ min	$P(W_q > 5)$	$P(W_q > 5) \leq 0.10$?
7	10.0964	4.6884	17989	6037	0.33559	No > 10%
8	2.2882	1.9761	18463	2500	0.13541	No > 10%

$P(W_q > 5) = \frac{\text{\# of } W_q > 5 \text{ min}}{\text{Total Customers}}$ is the probability that the expected time customers wait before beginning service is greater than 5 minutes

The minimum number of servers recommended to meet the goal is $s > 8$ since
$P(W_q > 5) = \frac{2500}{18463} = 0.13541 > 10\%$.

Please note that these numbers come from a simulation with 100 Samples. Since we were informed that there is room for 8 registers, and we need more than 8 registers, when working with trainees, this goal is not possible. Here is a graph showing the probability that some cashiers are busy but there is no line during each of these scenarios for 7 and 8 servers.



Conclusion
How Many Servers?

When the service time is and average of 6.5 minutes, we see that at 7 servers, $P(W_q > 5) \leq .10$. However, when the service time is an average of 8 minutes, we see that at 9 servers, $P(W_q > 5) \leq .10$. My recommendation is when you have several new servers with an average service time of 8 minutes, have 9 cashiers present, yet when you have servers averaging a 6.5 minutes service time, you can have 7 servers working. Since you can expect to have periods during the holidays where you need to hire trainees with an 8 minute service time average, I suggest building 9 stations to prepare for this. Unfortunately, due to the constraints of this problem (given when we were told that there is only room for 8 stations), it appears that having 9 cashiers is not possible. Perhaps it is just best to not have an 8 minute average service time if you want to meet this goal.

Average Service Time	Number of Servers	$P(W_q > 5$ minutes $)$
6.5 minutes	6 servers	27.51%
6.5 minutes	7 servers	6.46%
6.5 minutes	8 servers	2.54%
8 minutes	7 servers	33.56%
8 minutes	8 servers	13.54%
8 minutes	9 servers	4.57%